

Rochester Institute of Technology
Department of Computer Engineering
CMPE-630 Digital IC Design
Spring 2025
Laboratory Exercise 7
Sequential Logic: Designing a D Flip-Flop

Due: 03/28/2025

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1 Introduction

The goal of this lab exercises is for students to finish building up their library and adding a D Flip-Flop. Using the material learned in class about sequential logic, they will be able to apply analysis and behavior of this design.

There is **no lab report for this lab**. Complete the datasheet from section 5. See section 6 for required signoff items. **Submit the datasheet and the signoff sheet to lab dropbox.**

Keep the following in mind as you work through this lab.

- **NEVER** COPY CADENCE LIBRARY FILES FROM THE COMMAND LINE OR FILE MANAGER
- IF ANY OF YOUR CADENCE FILES ARE LOCKED, CLOSE THE TOOL AND EXECUTE THE FOLLOWING COMMAND:

```
/classes/ee620/bin/unlock_cadence.csh
```

- **Hotkeys:** You should learn how to use the hotkeys for better productivity. You can access a list of hotkeys (also known as bindkey) from CIW (the first window that pops up after you launch virtuoso). In CIW, select the menu option:

```
Options → Bindkey
```

Select a category (e.g. **layout**) and a list of current hotkeys (bindkeys) will be shown for it.

REMINDERS:

- Always start the Cadence tools using the **virtuoso &** command from your cmpe630 directory and NOT from your home directory.
- Always properly close the Cadence tools before you log off, otherwise you WILL lose your work and/or potentially lock your cells and not be able to access them!
- DO NOT delete any Cadence files from command line or file manager!

2 Procedure

Your D Flip-Flop (DFF) will be modeled after the circuit shown in Figure 1. D as your input, ϕ as your clock and the complementary clock. Then your output Q. To avoid having two clock inputs, add an extra inverter to ϕ and route the output to any complementary ϕ .

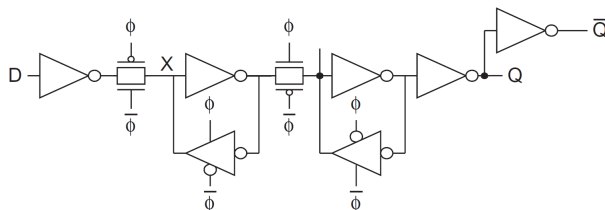


Figure 1: D Flip-Flop Logic Gate Diagram

Note that there is a new gate, this is a $C^2MOS - Latch$, you will see a diagram of its corresponding transistor schematic in Figure 2. You may find more information on these in page 393 of the course textbook. (You might find our DFF design to be similar to the textbook except that we have a change on where the inverter that makes our complementary Q is connected.)

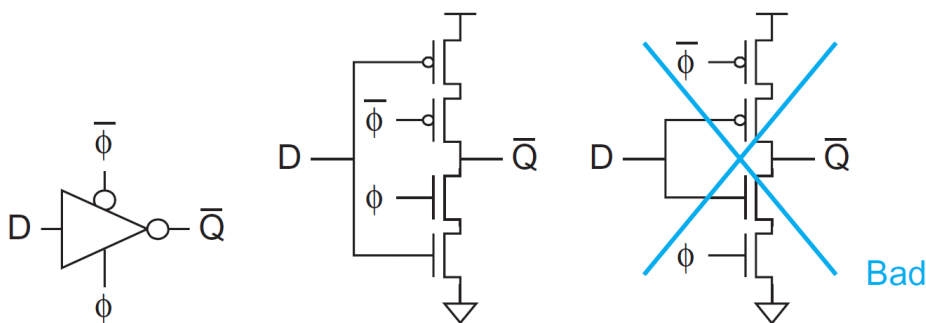


Figure 2: C^2MOS Latch Transistor Form

Synopsis: For this lab, you will design the schematic, symbol, and layout of the DFF. Create separate cells the D Flip-Flop and name it "FML" _DFFX1. Follow the same procedure you learned in previous labs to create the schematic, testbench and simulations. After you have all the schematics, create the layouts. Perform DRC, LVS and RC RCX extractions and simulate them. Take a look at Section 6 for the required signoff items. See next page for recommended design flow.

YOU MUST FOLLOW THE LAYOUT DESIGN GUIDELINES PROVIDED IN LAB5! Please refer to Lab5 for those guidelines. As mentioned there, we will not be able to provide signoff if your DFF layout does not meet all of the guidelines and rules.

Be mindful of the design rules and try your best to use the minimum measurements since you have to fit in a lot of stuff in a constrained space (height wise). Remember that you can change the

width as you go along by clicking on your template cell and pressing ‘Q’ on your keyboard.

Recommended flow for designing the DFF:

1. Draw the CMOS schematic (in paper) and stick diagram. This will help you visualize the general structure of the DFF design in both schematic and layout level
2. Create a separate cell for the DFF, and name it "*FML_DFFX1*" , note that FML stands for First, Middle, and Last initial of your name
3. Create a new schematic view, and design your DFF schematic for the newly created cell. Note that the latch transistors are slightly different so make sure to look at it closer to determine where the right inputs belong
4. Create a symbol for your DFF and perform testings listed in Section 4 to ensure the correctness of your design
5. Create a layout for your DFF and performs successful DRC, LVS, and QRC
6. Rerun tests with extracted circuits to find post-layout timings

Good luck, be patience, and more importantly have fun!

3 Measuring Propagation Delay

Propagation delay is measured a little differently so it's visible to see the changes on the waveform. For this, we will be using hot keys! Bring up the first waveform and estimate the midpoint at the first falling edge. Press **m** on your keyboard and an instantaneous marker will appear. Since this isn't accurate, we will change the value. Either press "q" key for the properties or right click on your new measurement and click on **Point Maker Properties**.

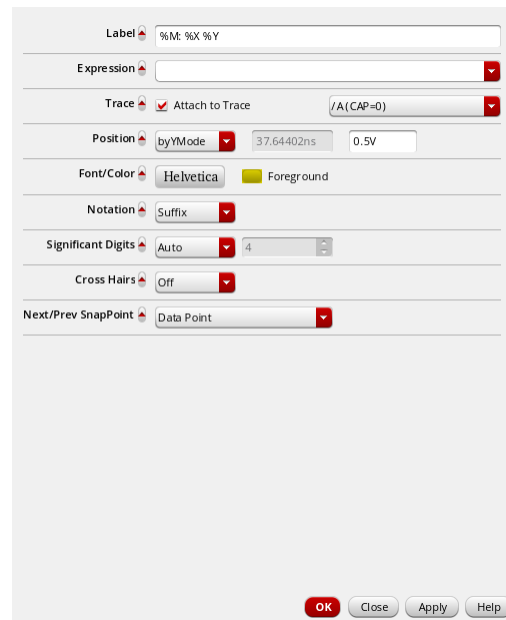


Figure 3: Point Properties

Here Go to "Position" and change it to "YMode" and change the value to be 0.5V so we get an accurate midpoint. Click Apply and Ok or the Enter key.

Now press the key "d" on your keyboard and estimate the midpoint of the output you would like to calculate delay with. Similarly make sure to change the property to be exactly 0.5V. Fig. 4 is a dummy example of how a delay should look like.

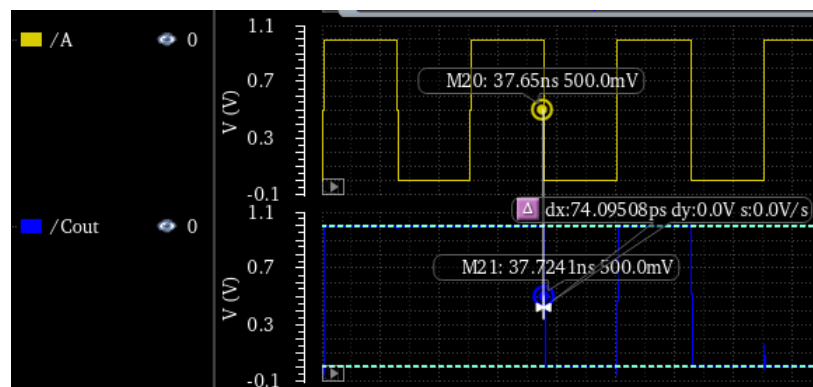


Figure 4: Delay example

4 Testing

To show the functionality of your flip flop you will provide a normal waveform (i.e. functional simulation waveform) and you will also find the Set-up time, hold time ¹ and the clock-to-Q (i.e. propagation from clk to q) delay. Find one case that pass and one that fail for each one of set up time and hold time and one measurement of clock-to-Q delay. When you make the ADE XL test this is the following set up:

- CAP : 100f
- Temperature : 125c
- Test Editor Inputs for Setup/Hold times:
 - Voltage 1: 1V
 - Voltage 2: 0V
 - Period: (ϕ 50n) / (D 100n)
 - Rise time: 1ps
 - Fall time: 1ps
 - Delay time: 25n for both ϕ and D. However, you should change and experiment other delays with D to find out setup/told time.
- Test Editor Inputs for Clock-to-Q Delay
 - Voltage 1: 1V
 - Voltage 2: 0V
 - Period: 50n
 - Rise time: 100ps
 - Fall time: 100ps
 - Delay time: 25n
- Test Editor Global Sources is the same as usual.

Note on setup and holdtime: Recall the definition of setup and hold time, which involved some minimum time the input has to be stable before and after the clock edge. You will use the brute force method, which is to try out different delay time value until one of the times is violated.

¹In the case that hold time is negative, instead of reporting an actual value explain the followings in the comment section of the datasheet: 1) why it is negative? 2) what does negative holdtime indicate?

5 DFF Datasheet

Complete the data sheet for this DFF design. The template specific to the DFF is provided in MyCourses.

6 Sign-off Sheet

You don't need to have the 'Worksheet' sign-off before the lab period, it will be graded after it's turned in.

Table 1: Small Layout Design Library

Students Name:		Section:	
Demo	Point Value	Points Earned	Date
DRC Check	25		
LVS Check	25		
Parasitic RC RCX	10		
Datasheet	40		
Total Points	100		